Sheaf-based geometric deep learning for p53 signalling response prediction

IFOM (Buffa / Tangherloni) x Cambridge (Lio) - flagship preprint target, M0-M5 sprint · Gabriele Bambini · draft v1 · 2026-08-26

HYPOTHESIS

Plain message-passing GNNs oversmooth and miss directionality on cell-level signalling graphs. Constructing a cellular sheaf over pathway topology (one stalk per protein complex edge, grounded in expression state) should improve prediction of p53-rescue fitness phenotypes and yield calibrated, TP53-status-aware uncertainty bands for wet-lab triage.

AIMS

A1 signalling-sheaf construction from DepMap + pathway topology | A2 benchmark vs GNN baselines on p53-rescue fitness | A3 TP53-stratified conformal uncertainty -> wet-lab triage list.

METHODS

Data: DepMap expression + CRISPR fitness labels; TCGA stratification; Reactome/STRING topology. Models: neural sheaf diffusion vs GCN/GAT/graph-transformer baselines under identical splits; conformal wrappers for calibrated sets. Validation: rank correlation with independent screens; pathway enrichment of top predictions reviewed with Buffa lab; HPC runs on IFOM cluster (Tangherloni).

EXPECTED OUTPUTS

bioRxiv preprint (M4 milestone); curated sheaf-dataset release; wet-lab triage shortlist co-signed by both labs.

BIBLIOGRAPHY & CHAIN OF VERIFICATION (4/11 SOURCES LIVE ON 2026-08-26)

Claim used	Source	Link	CoV
Sheaf Neural Networks introduce cellular-sheaf message passing	Hansen & Gebhart, arXiv 2020	https://arxiv.org/abs/2012.03119	◎ **unverified**
Neural sheaf diffusion targets heterophily/oversmoothing	Bodnar et al., CVPR 2022	https://arxiv.org/abs/2202.04579	◎ **unverified**
GCN baseline formalism	Kipf & Welling, ICLR 2017	https://arxiv.org/abs/1609.02907	**✓ HTTP 200**
GAT attention baseline	Velickovic et al., ICLR 2018	https://arxiv.org/abs/1710.10903	**✓ HTTP 200**

p53 pathway centrality motivates target phenotype	Vogelstein et al., Nature Medicine 2004	https://doi.org/10.1038/nm1087	◎ **unverified**
DepMap expression correction enables pan-cancer features	Meyers et al., Nat Genet 2017	https://doi.org/10.1038/ng.3914	◎ **unverified**
Dependency Map provides CRISPR fitness labels	Tsherniak et al., Cell 2017	https://doi.org/10.1016/j.cell.2017.06.010	◎ **unverified**
TP53 mutation spectrum for stratification	IARC TP53 Database	https://p53.fr/	◎ **unverified**
Conformal prediction gives distribution-free bands	Angelopoulos & Bates, 2021	https://arxiv.org/abs/2107.07511	✓ **HTTP 200**
[CoV] DepMap portal reachable, research terms available	data source check	https://depmap.org/	✓ **HTTP 200**
[CoV] STRING API reachable for edge construction	data source check	https://string-db.org/	◎ **unverified**

Chain-of-verification method: every claim row was probed automatically at build time (2026-08-26); ✓ = endpoint answered HTTP 200 during generation, ◎ = probe failed or blocked (verify manually before citing in applications). Draft generated by the PhD Funding Command Center build.